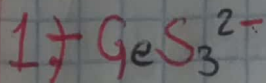
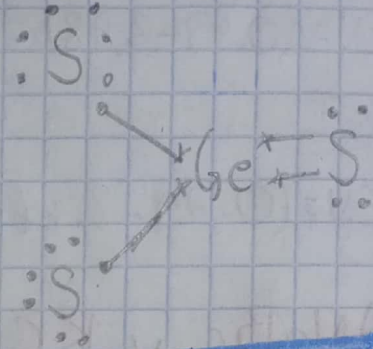
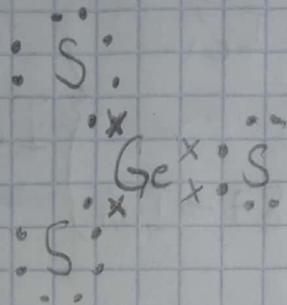
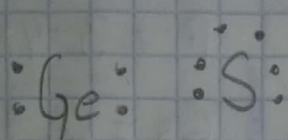


Exámen

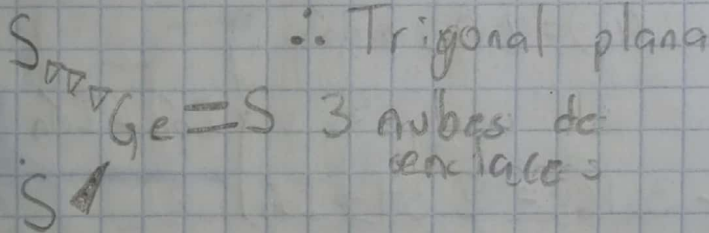


a) Estructura de Lewis $e^- = 24 e^-$

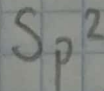
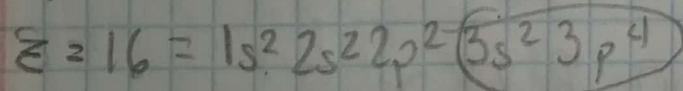


3 y 1.7

b) Geometría molecular



c) Hibridación del átomo central



Nombre:

Día

Mes

Año

Folio

Tema:

2.~

a) LiF y CsI

LiF

$$\Delta EN = (0.98 - 3.98) = 3 \rightarrow \text{Ión}$$

CsI

$$\Delta EN = (0.79 - 2.66) = 1.87 \rightarrow \text{Ión}$$

Ión - Ión

Electronegatividad

$$Li = 0.98$$

$$F = 3.98$$

$$Cs = 0.79$$

$$I = 2.66$$

$$Na = 0.93$$

$$Br = 2.96$$

$$K = 0.82$$

$$Cl = 3.16$$

$$H = 2.2$$

b) NaBr y KCl

NaBr

$$\Delta EN = (0.93 - 2.96) = 2.03 \rightarrow \text{Ión}$$

KCl

$$\Delta EN = (0.82 - 3.16) = 2.34 \rightarrow \text{Ión}$$

Ión - Ión

c) ClF y HI

ClF

$$\Delta EN = (3.16 - 3.98) = 0.82 \rightarrow \text{Dipolo}$$

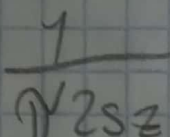
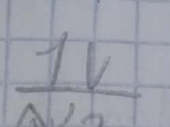
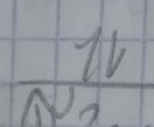
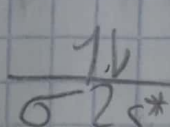
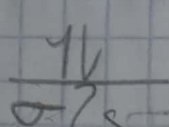
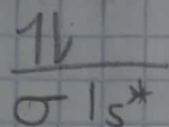
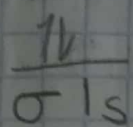
HI

$$\Delta EN = (2.2 - 2.66) = 0.46 \rightarrow \text{Dipolo}$$

Dipolo - Dipolo

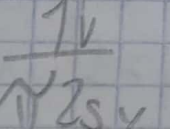
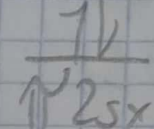
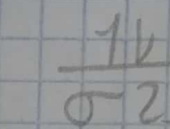
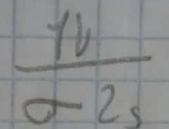
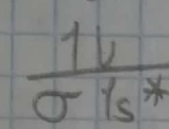
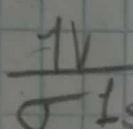
3.-

$$CN = 13e^-$$



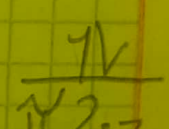
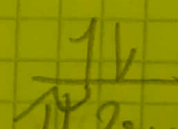
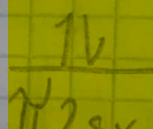
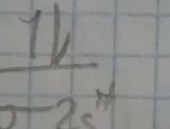
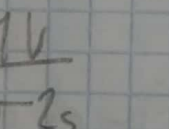
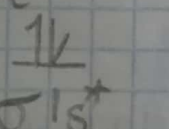
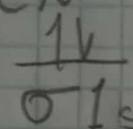
→ Paramagnética $\rightarrow O_E = \frac{1}{2}(9-7) = \underline{1}$

$$CN^{1+} = 12e^-$$



$$O_E = \frac{1}{2}(8-4) = \underline{2} \rightarrow \text{Diamagnética}$$

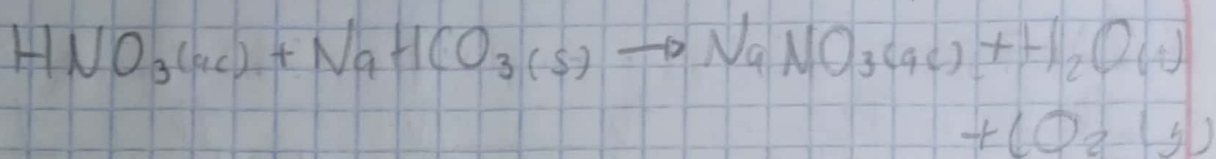
$$CN^{1-} = 14e^-$$



$$O_E = \frac{1}{2}(10-4) = \underline{3} \rightarrow \text{Diamagnética}$$

$$\underline{CN^{1+} < CN < CN^{1-}}$$

4.-



Balanceo

$$\text{HNO}_3 = 63.008$$

$$\text{H} = 2 \quad \text{H} = 2$$

$$\text{N} = 1 \quad \text{N} = 1$$

$$\text{O} = 6 \quad \text{O} = 6$$

$$\text{Na} = 1 \quad \text{Na} = 1$$

$$\text{C} = 1 \quad \text{C} = 1$$

$$550 \text{ mL} \left(\frac{7.5 \text{ g HNO}_3}{100 \text{ mL}} \right) \left(\frac{1 \text{ mol HNO}_3}{63.008 \text{ g HNO}_3} \right) = 0.38839512 \text{ mol HNO}_3$$

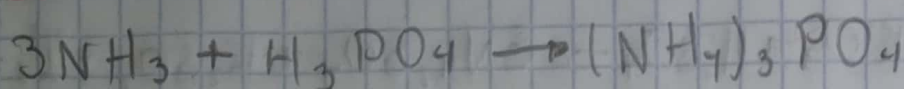
$$3.4328 \times 10^{23} \text{ mol} \left(\frac{1 \text{ mol}}{6.02214 \times 10^{23}} \right) = 0.570029922 \text{ mol NaHCO}_3$$

$$n = \frac{P \cdot V}{R \cdot T} = \frac{(1.093 \text{ atm})(7.7 \text{ L})}{(0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}})(294.15 \text{ K})} = 0.350 \text{ mol CO}_2$$

$$\text{Rendimiento CO}_2 = 0.35 \text{ mol CO}_2 \quad 0.388 \text{ mol HNO}_3 \left(\frac{1 \text{ mol CO}_2}{1 \text{ mol HNO}_3} \right) = 0.388 \text{ mol HNO}_3$$

$$\text{Rend \%} = \left(\frac{0.35 \text{ mol CO}_2}{0.388 \text{ mol HNO}_3} \right) \times 100 = 90.206\%$$

S:-



$$N = 3$$

$$N = 3$$

$$H = 9 + 3 = 12$$

$$H = 12$$

$$P = 1$$

$$P = 1$$

$$O = 4$$

$$O = 4$$

$$200 \text{ mL dis} \left(\frac{85 \text{ H}_3\text{PO}_4}{100 \text{ mL dis}} \right) \left(\frac{1 \text{ mol H}_3\text{PO}_4}{97.994 \text{ H}_3\text{PO}_4} \right) = 1.7348 \text{ mol H}_3\text{PO}_4$$

$$V = \frac{n \cdot R \cdot T}{P} \quad V = \frac{3 (0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})}{0.76 \text{ atm}} = 96.80$$

$$V = 96.8064$$

$$V = \frac{m}{MM} \frac{R T}{P} = \frac{(110 \text{ g}) (0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}) (298.15 \text{ K})}{(51.102) (0.76 \text{ atm})}$$

$$V_{\text{NH}_3} = 107.15 \text{ [L]}$$

Nombre:

Día

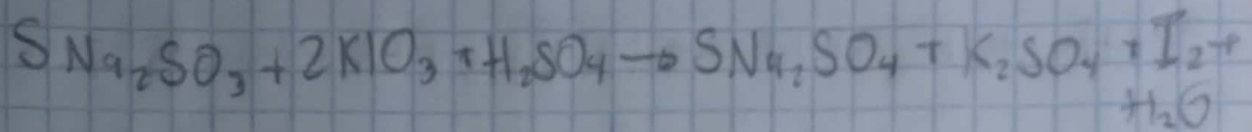
Mes

Año

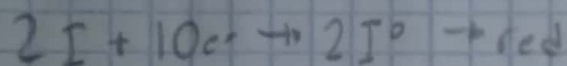
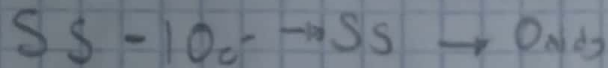
Folio

Tema:

6.-



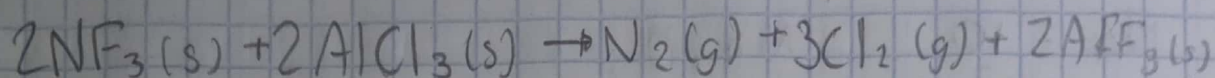
Reacción oxidación - reducción



Na_2SO_3 es un agente reductor, KIO_3 es de oxidación

I_2 es el producto de $5\text{Na}_2\text{SO}_3 + 2\text{KIO}_3 + \text{H}_2\text{SO}_4$,
y que, a fin de cuentas, da 2

7.-



$$70\text{g NF}_3 \left(\frac{1\text{ mol NF}_3}{71.0019\text{g NF}_3} \right) = 0.9859\text{ mol NF}_3$$

$$140\text{g AlCl}_3 \left(\frac{1\text{ mol AlCl}_3}{133.3408\text{g AlCl}_3} \right) = 1.0499\text{ mol AlCl}_3$$

$$1.0499\text{ mol AlCl}_3 \left(\frac{2\text{ mol NF}_3}{2\text{ mol AlCl}_3} \right) = 1.0499\text{ mol NF}_3$$

$$0.9859\text{ mol NF}_3 \left(\frac{2\text{ mol AlCl}_3}{2\text{ mol NF}_3} \right) = 0.9859\text{ mol AlCl}_3$$

$$0.9859\text{ mol NF}_3 \left(\frac{2\text{ mol AlF}_3}{2\text{ mol NF}_3} \right) = 0.9859\text{ mol AlF}_3$$

$$\% \text{ rend} = \left(\frac{0.8972\text{ mol AlF}_3}{0.9859\text{ mol AlF}_3} \right) \times 100 = 91.0031\%$$